

Software Architecture

Exercise – Architecture Significance

BSc



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Exercise Opening

Overview

In this exercise, **you will be given requirements** for a **new system** that you are **expected to design** and develop.

For **each requirement**, you need to **consider** how **architecture-significant** it is.

A **requirement is significant** if the **software design** to which the **requirement is likely to lead** is a **design** that is **difficult to dismantle, change** or **adapt** should the **requirement change** in the **future**.

Exercise Opening

Overview

You will be organised into **six working groups** and find **requirements** for a **fictitious software system** on the **following slides** (each group receives **4 requirements** to be assessed).

Your task is to ...

1. **Briefly assess** for each **requirement** whether it will **significantly shape** the **architecture** of an appropriate system.
2. For **those requirements** that **lead to far-reaching design decisions** (i.e., architecture-significant design decisions), **sketch** the **resulting design**.
 - a. **Sketch out** the **coarse-grained components** that **play a role** in your **design** (question: **which components are needed?**).
 - b. **Select** a **component** from (a) and **suggest software architecture approaches** at the **next level of detail** for this **component** (question: **which software design approaches** do I **propose** for a **selected component** from (a)?).
3. **Explain** to what **extent** the **design decisions** you noted would be **difficult to adapt or reverse** if the requirements changed.

Work out your requirement assessments based on the following template: [4-BSc_SWA_Conditions_Significance_Template.docx](#).

Exercise Opening

Overview

List of requirements and student groups to which they are assigned.

No.	Requirement	Working Group	Significance
1	The system must be highly available 24/7.	Group 1	
2	The system should display the company logo at the top left of each page.	Group 1	
3	Users must log in with a username and password.	Group 1	
4	All system functions must also be available via a mobile app.	Group 1	
5	The system must process personal data in accordance with the GDPR.	Group 2	
6	The data entry page should include a calendar component for date fields.	Group 2	
7	The application must be available in German and English.	Group 2	
8	All business processes must be traceable and auditable for audit purposes.	Group 2	
9	The system must be integrated into the company's existing SAP environment.	Group 3	
10	The background colour of the start page should be blue.	Group 3	
11	The app should also be usable in landscape mode.	Group 3	
12	The solution should be cloud-native (e.g. on AWS).	Group 3	
13	The system should be configurable by business users without developer support.	Group 4	
14	Users should be able to sort data records by columns.	Group 4	
15	Password policies should require at least 8 characters and one capital letter.	Group 4	
16	The system must be fully multi-client capable.	Group 4	

Exercise Opening

Overview

List of requirements and student groups to which they are assigned.

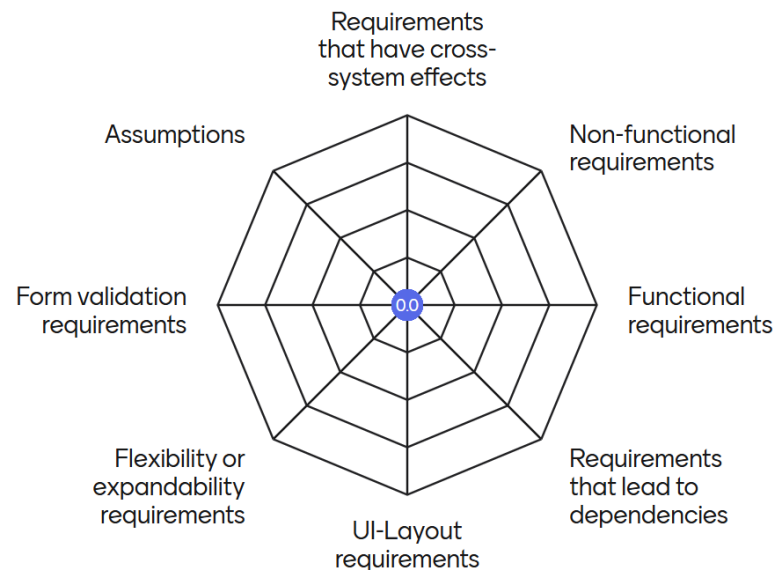
No.	Requirement	Working Group	Significance
17	The system must meet real-time requirements (e.g. responses within 50 ms).	Group 5	
18	The system should automatically log out after 15 minutes of inactivity.	Group 5	
19	Input fields should perform validations (e.g. email format).	Group 5	
20	The system must be expandable by third parties (e.g. partner companies) via publicly documented interfaces.	Group 5	
21	The system must support machine learning (e.g. by training and using models based on usage data).	Group 6	
22	The application should offer a print function for invoices.	Group 6	
23	The system should allow users to reset their passwords themselves.	Group 6	
24	The system must support rule-based decision logic that can be adapted by specialists at runtime.	Group 6	

Exercise Opening

Overview

Let us **discuss** the **following questions** in order to **reflect** on the **significance assessments** that we have performed:

1. Which were **common characteristics** of **requirements** that were **architecture significant**?
2. What is the **extent** to which the following **types of requirements** tend to be **architecture-significant** (answers are provided in the respective Mentimeter)?



Exercise Opening

Overview

Let us **discuss** the **following questions** in order to **reflect** on the **significance assessments** that we have performed:

1. Are there **requirements** that **do not initially appear** to be **relevant** to the **architecture** but **become so later** on? Let us have a look at a few examples:
 - a. Modifiability in the user interface (Example: “Specialist users should be able to customise UI fields”. Initial assessment: “Only a UI issue, low relevance”).
 - b. Internationalization (Example: “Texts should be displayed in German and English”. Initial assessment: Localisation via UI labels).
 - c. Mobile use (Example: “The app should also work on tablets”. Initial assessment: UI-only issue).
 - d. Role and rights model (Example: “Admin and user rights must be differentiated”. Initial assessment: Implementation in the front end, simple roles).
 - e. Export function as ‘simple CSV’ (Example: “Data should be exportable”. Initial assessment: Pure formatting function).
 - f. Reporting/statistics (Example: “A reporting module would be nice”. Initial assessment: Can be added later).

Wrap-Up

- Is provided in **4-BSc_SWA_Conditions_Significance-Solution.pdf**

Questions

